Use-Case Flow Specification

Use Case Name	Generate Immutable Audit Export
Primary Actor	Compliance Officer
Secondary Actor(s)	Security Auditor (consumes the export); Cryptographic Signing Service (system)
Description	Enables a Compliance Officer to produce a tamper-evident, cryptographically verifiable export of masked audit log records for a specified scope and date range, satisfying regulatory audit and e-discovery requests.
Preconditions	• The Compliance Officer is authenticated and has an active session with export-generation privileges. • At least one ingested audit log record, already PII-masked per UC-02 (Mask PII Fields), exists within the requested date range and scope. • The platform's cryptographic signing service / HSM (Hardware Security Module) is provisioned and reachable.
Postconditions	Success guarantee: • An immutable, cryptographically signed audit export bundle is persisted in the export ledger, tagged with its SHA-256 integrity hash, digital signature, and generation timestamp. • The export is available for download by the requesting Compliance Officer and, where authorized, the Security Auditor. • A new audit trail entry records who generated the export, when, and its exact scope (for non-repudiation). Minimal guarantee (failure case): • No partial or unsigned export bundle is persisted; a failure event is logged for troubleshooting.
Main Success Scenario	1. Compliance Officer selects "Generate Audit Export" and specifies the date range, log source(s), and export format (CSV/JSON). 2. System verifies the Compliance Officer's role and export permissions for the requested scope. 3. System retrieves the matching, already PII-masked audit log records from storage. 4. System compiles the retrieved records into a single export bundle in the requested format. 5. System computes a SHA-256 integrity hash over the finalized export bundle. 6. System sends the bundle and hash to the Cryptographic Signing Service, which digitally signs them and returns a signed manifest with a timestamp. 7. System stores the export bundle together with its hash, signature, and timestamp in the immutable export ledger (write-once storage). 8. System notifies the Compliance Officer that the export completed successfully and provides a secure download link. 9. Compliance Officer downloads the signed audit export. 10. Use case ends successfully.
Alternate Flow (6a. Signing Service Unavailable)	6a1. System detects that the Cryptographic Signing Service / HSM cannot be reached or returns an error while attempting to sign the export bundle. 6a2. System automatically retries the signing request up to 2 additional times with exponential backoff. 6a3. If signing still fails after 2 retries, System aborts the export, discards the unsigned bundle (it is never written to the export ledger), and logs a signing-failure event. 6a4. System notifies the Compliance Officer that export generation failed and that they may retry once the signing service is restored. 6a5. Use case ends without producing an export; flow returns to step 1 if the Compliance Officer chooses to retry.